Essential TypeScript Topics for Learning a Playwright Automation Framework

Goal

Learn enough TypeScript to confidently understand, debug, and contribute to a Playwright automation framework used in an organization. The topics below are prioritized for QA automation rather than frontend development.

Priority 1 – Must Know Before Reading the Framework

• Variables and type annotations

• Type inference

• Primitive types

• Arrays and tuples

• Objects

• Functions (optional/default/rest parameters, return types)

• Arrow functions

• Template literals

• Destructuring

• let, const, scope

Priority 2 – Essential TypeScript Concepts

• Interfaces

• Type aliases

• Union types

• Literal types

• Optional properties

• Readonly properties

• Enums (basic understanding)

Priority 3 – Object-Oriented Programming

• Classes

• Constructors

• Access modifiers (public/private/protected)

• Readonly members

• Inheritance

• Abstract classes (basic)

Priority 4 – Modules & Project Structure

• import/export

- Named vs default exports
- File/module organization
- tsconfig.json (basic understanding)
- Path aliases (if used)

Priority 5 – Async Programming

- Promises
- async/await
- Error handling with try/catch
- Promise.all (basic)

Priority 6 – Commonly Used in Playwright

- Generics (basic)
- Type assertions
- Type narrowing
- Built-in utility types (Partial, Pick, Omit, Record)

Priority 7 – Playwright-Specific Concepts

- Typing Page Objects
- Typing fixtures
- Playwright Test object model
- Configuration files
- Environment variables
- Reusable helper functions

Can Learn Later

- Decorators
- Namespaces
- Declaration merging
- Advanced generics
- Mixins

Suggested Learning Order

1. Basics → 2. Functions → 3. Interfaces & Type Aliases → 4. Classes → 5. Modules → 6. Async/Await → 7. Generics → 8. Utility Types → 9. Playwright-specific patterns.

Estimated Effort

With prior JavaScript knowledge, the Priority 1–6 topics can typically be learned in 2–3 weeks of focused study (1–2 hours/day).